

Notes on Template Games and Differential Logic

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1 General Remarks

The results from [FGHW08] introduce the bicategorical interpretation of linear logic based on distributors and generalised species. The idea is that every formula A of linear logic is interpreted as a small category $\llbracket A \rrbracket$ and every derivation tree

$$\frac{\begin{array}{c} \pi \\ \dots \end{array}}{A_1, \dots, A_n \vdash A}$$

is interpreted as a distributor (or a profunctor)

$$\llbracket \pi \rrbracket : \llbracket A_1 \rrbracket \otimes \dots \otimes \llbracket A_n \rrbracket \not\rightarrow \llbracket A \rrbracket.$$

Recall that a distributor $M : A \not\rightarrow B$ is defined as a functor $M : A \times B^{op} \rightarrow \mathbf{Set}$. The category **Dist** of small categories and distributors is symmetric monoidal with the tensor product defined by the direct product of categories in **Cat**, the category of small categories and functors, and the unit defined by the terminal (a single-element) category. **Dist** is also *-autonomous (and, moreover, compact closed) with linear negation defined as $A \mapsto A^{op}$. We thus have the following isomorphism

$$\mathbf{Dist}(A \otimes B, C) \cong \mathbf{Dist}(B, A^{op} \otimes C)$$

natural A , B and C , providing the curriification in **Dist**.

The exponential modality $A \mapsto !A$ is interpreted using the 2-monad **Sym** : **Cat** $\rightarrow$ **Cat** sending each small category $\mathcal{C}$ to the free symmetric monoidal category generated by $\mathcal{C}$. **Sym**($\mathcal{C}$) has objects defined as finite sequences $A_1 \dots A_n$ for objects from $\mathcal{C}$ for where $A_i \in \mathcal{C}$ for $1 \leq i \leq n$, and morphisms

$$(\sigma, f_1 \dots f_n) : A_1 \dots A_n \rightarrow B_1 \dots B_n$$

defined as pairs $(\sigma, f_1 \dots f_n)$ consists of a permutation $\sigma \in \mathcal{S}_n$ on the set of n -elements along with a sequence of morphisms from $\mathcal{C}$ for $1 \leq k \leq n$

$$f_k : A_k \rightarrow B_{\sigma(k)}.$$

The category $\mathbf{Sym}(\mathcal{C})$, as strict monoidal, comes with a pair of functors $\otimes : \mathbf{Sym}(\mathcal{C}) \times \mathbf{Sym}(\mathcal{C}) \rightarrow \mathbf{Sym}(\mathcal{C})$ and $I_A : \mathbf{1} \rightarrow \mathbf{Sym}(\mathcal{C})$ defined as the concatenation and the empty sequence respectively. $\mathbf{Sym}(\mathcal{C})$ with those two functors also forms a monoid in $\mathbf{Cat}$, and such a monoid is commutative up to isomorphism γ (the symmetry of the monoidal category):

$$\begin{array}{ccc} \mathbf{Sym}(\mathcal{C}) \times \mathbf{Sym}(\mathcal{C}) & \xrightarrow{(21)} & \mathbf{Sym}(\mathcal{C}) \times \mathbf{Sym}(\mathcal{C}) \\ & \searrow \otimes \quad \xrightarrow{\alpha} \quad \swarrow \otimes & \\ & \mathbf{Sym}(\mathcal{C}) & \end{array}$$

where (21) is the symmetry of the cartesian category $\mathbf{Sym}(\mathcal{C}) \times \mathbf{Sym}(\mathcal{C})$. The exponential modality $A \mapsto !A$ is, thus, interpreted in $\mathbf{Dist}$ by turning $\mathbf{Sym}$ into a 2-comonad

$$\mathbf{Sym} : \mathbf{Dist} \rightarrow \mathbf{Dist}$$

using the distributivity law between $\mathbf{Sym}$ and the presheaf construction in $\mathbf{Cat}$. Further, every functor $F : \mathcal{C} \rightarrow \mathcal{D}$ induces an adjoint pair $L_F \dashv R_F$ of distributors $L_F : A \nrightarrow B$ and $R_F : B \nrightarrow A$ in the bicategory $\mathbf{Dist}$, where the distributors L_F and R_F are given as the following presheaves

$$\begin{aligned} L_F(B, A) &= \mathcal{D}(FB, A) : B^{op} \times A \rightarrow \mathbf{Set} \\ R_F(A, B) &= \mathcal{D}(A, FB) : A^{op} \times B \rightarrow \mathbf{Set} \end{aligned}$$

The comonoid structure $(\mathbf{Sym}(\mathcal{C}), d_{\mathcal{C}}, e_{\mathcal{C}})$ of the exponential modality $\mathbf{Sym}$ is given by the following distributors

$$\begin{aligned} d_{\mathcal{C}} &= R_{\otimes_{\mathcal{C}}} : \mathbf{Sym}(\mathcal{C}) \nrightarrow \mathbf{Sym}(\mathcal{C}) \otimes \mathbf{Sym}(\mathcal{C}) \\ e_{\mathcal{C}} &= R_{I_{\mathcal{C}}} : \mathbf{Sym}(\mathcal{C}) \nrightarrow \mathbf{1} \end{aligned}$$

right adjoint to $\otimes_{\mathcal{C}}$ and $I_{\mathcal{C}}$ respectively. Note that $(\mathbf{Sym}(\mathcal{C}), d_{\mathcal{C}}, e_{\mathcal{C}})$ is not commutative, but only symmetric, so it is commutative only up to isomorphism:

$$\begin{array}{ccc} & \mathbf{Sym}(\mathcal{C}) & \\ d_{\mathcal{C}} \swarrow & \xrightarrow{\alpha} & \searrow d_{\mathcal{C}} \\ \mathbf{Sym}(\mathcal{C}) \otimes \mathbf{Sym}(\mathcal{C}) & \xrightarrow{(21)} & \mathbf{Sym}(\mathcal{C}) \otimes \mathbf{Sym}(\mathcal{C}) \end{array}$$

where γ is the symmetry of $\mathbf{Sym}(\mathcal{C}) \otimes \mathbf{Sym}(\mathcal{C})$, which is a distributor in $\mathbf{Dist}$ as well.

The necessity to weaken the commutativity condition to an isomorphism is similar to the phenomenon of the Seely isomorphism in linear logic:

$$!(A \& B) \cong !A \otimes !B$$

One can interpret this equation of linear logic as a distributor with the concatenation functor:

$$\text{concat}_{A,B} : \mathbf{Sym}(A) \times \mathbf{Sym}(B) \rightarrow \mathbf{Sym}(A + B)$$

that takes sequences $u = a_1 \dots a_p$ and $v = b_1 \dots b_q$ of objects from A and B respectively concatenates them into the sequence of objects from $A + B$:

$$\text{concat}_{A,B}(u, v) = a_1 \dots a_p \cdot b_1 \dots b_q$$

The distributor interpreting the Seely isomorphism

$$\mathbf{Sym}(A \& B) \not\cong \mathbf{Sym}(A \otimes B)$$

is defined as the right adjoint of the concatenation functor where $A \& B$ stands for $A + B$ for the disjoint union of A and B . Besides, $\text{concat}_{A,B}$ is equivalence of categories injective on objects, but it is not an isomorphism. Thus we replace here the Seely isomorphism with the Seely equivalence.

2 On Models of Differential Linear Logic

A good introduction to differential linear logic is [Ehr18]. If we interpret the exponential modality as $A \mapsto \mathbf{Sym}(A)$, then one can see that the model of distributors is also a model of differential linear logic since the exponential modality $\mathbf{Sym}(A)$ comes with the monoid structure in **Dist**:

- $m_A = L_{\otimes A} : \mathbf{Sym}(A) \otimes \mathbf{Sym}(A) \rightarrow \mathbf{Sym}(A)$,
- $u_A = L_{I_A} : \mathbb{1} \rightarrow \mathbf{Sym}(A)$.

whose multiplication and unit are the left adjoint distributors associated to the monoid structure of $\mathbf{Sym}(A)$ in **Cat**. Note that the monoid $(\mathbf{Sym}(A), m_A, u_A)$ and the comonoid $(\mathbf{Sym}(A), d_A, e_A)$ are the left and right adjoint “avatars” of the monoid structure $(\mathbf{Sym}(A), \otimes_A, I_A)$ in **Cat**. The monoid and the comonoid structures on $\mathbf{Sym}(A)$ define a bialgebra, but the bialgebra structure is only up to an invertible natural transformation:

$$\begin{array}{ccccc}
 & & \mathbf{Sym}(A) & & \\
 & \nearrow m_A & \parallel & \searrow d_A & \\
 \mathbf{Sym}(A)^{\otimes 2} & & \downarrow \sim & & \mathbf{Sym}(A)^{\otimes 2} \\
 & \searrow d_A \otimes d_A & & \nearrow m_A \otimes m_A & \\
 & \mathbf{Sym}(A)^{\otimes 4} & \xrightarrow{(1324)} & \mathbf{Sym}(A)^{\otimes 4} &
 \end{array}$$

The explanation is that the disjoint sum $A + B$ of small categories A and B coincide with their cartesian sum $A \oplus B$ and their cartesian product $A \& B$. Therefore, every object A is equipped with a monoid and a comonoid structure:

- $\nabla_A : A \oplus A \not\rightarrow A$,
- $\nabla_A^0 : 0 \not\rightarrow A$,
- $\Delta_A : A \not\rightarrow A \oplus A$,
- $\Delta_A^0 : A \not\rightarrow 0$.

defined as the left and right adjoint distributors to the canonical functors $A + A \rightarrow A$ and $0 \rightarrow A$, where 0 is the initial category in **Cat**. The multiplication ∇_A and the comultiplication Δ_A are equipped with a bialgebra structure in **Dist**, up to an invertible natural transformation:

$$\begin{array}{ccccc}
 & & A & & \\
 & \nearrow \nabla_A & \parallel \sim & \nwarrow \Delta_A & \\
 A^{\oplus 2} & & & & A^{\oplus 2} \\
 & \searrow \Delta_A \oplus \Delta_A & & \nearrow \nabla_A \oplus \nabla_A & \\
 & A^{\oplus 4} & \xrightarrow{(1324)} & A^{\oplus 4} &
 \end{array}$$

The exponential modality $A \mapsto \mathbf{Sym}(A)$ along the family of Seelye equivalences

$$\mathbf{Sym}(A^{\oplus n}) \not\rightarrow (\mathbf{Sym}(A))^{\otimes n}$$

defines a lax and oplax symmetric monoidal bifunctor

$$\mathbf{Sym} : (\mathbf{Dist}, \oplus, 0) \rightarrow (\mathbf{Dist}, \otimes, \mathbb{1})$$

The differential in DiLL is interpreted the following way in the category of distributors. One starts with a family of functors

$$\eta_A : A \rightarrow \mathbf{Sym}(A)$$

defining the unit of the monad $A \mapsto \mathbf{Sym}(A)$. One can postcompose the functor with $\otimes_A$ to obtain the functor:

$$\text{append}_A : (a, v) \mapsto a \cdot v \in \mathbf{Sym}(A) : A \times \mathbf{Sym}(A) \rightarrow \mathbf{Sym}(A)$$

The differential is interpreted in **Dist** as the left adjoint distributor

$$\partial_A : A \otimes \mathbf{Sym}(A) \not\rightarrow \mathbf{Sym}(A)$$

associated to the append_A , whereas the dereliction and the codereliction morphisms:

- $\text{der}_A : \mathbf{Sym}(A) \not\rightarrow A$,
- $\text{coder}_A : A \not\rightarrow \mathbf{Sym}(A)$

are the left and right adjoint distributors associated to η_A .

The composite of the codereliction with the comultiplication

$$A \xrightarrow{\text{coder}_A} \mathbf{Sym}(A) \xrightarrow{d_A} \mathbf{Sym}(A) \otimes \mathbf{Sym}(A)$$

is equal to the disjoint union of the following two composites:

$$\begin{aligned} A &\xrightarrow{\text{coder}_A} \mathbf{Sym}(A) \xrightarrow{u_A \otimes 1_{\mathbf{Sym}(A)}} \mathbf{Sym}(A) \otimes \mathbf{Sym}(A) \\ A &\xrightarrow{\text{coder}_A} \mathbf{Sym}(A) \xrightarrow{1_{\mathbf{Sym}(A)} \otimes u_A} \mathbf{Sym}(A) \otimes \mathbf{Sym}(A) \end{aligned}$$

where the disjoint product of distributors $M, N : A \nrightarrow B$ is the distributor $M + N$ defined using the convolution product associated to the additive comonoid and monoid structures on A and B :

$$A \xrightarrow{\Delta_A} A \oplus A \xrightarrow{M+N} B \oplus B \xrightarrow{\nabla_B} B$$

or defined as the presheaf:

$$M + N : (b, a) \mapsto M(b, a) \sqcup N(b, a)$$

The property of the dereliction ensures that the Leibniz rule is satisfied, in the technical sense that the composite distributor

$$A \otimes \mathbf{Sym}(A) \xrightarrow{\partial_A} \mathbf{Sym}(A) \xrightarrow{d_A} \mathbf{Sym}(A) \otimes \mathbf{Sym}(A)$$

is isomorphic to the sum of the two distributors obtained by precomposing the distributor:

$$A \otimes \mathbf{Sym}(A) \xrightarrow{1_A \otimes d_A} A \otimes \mathbf{Sym}(A) \otimes \mathbf{Sym}(A)$$

with the two distributors represented below:

$$\begin{array}{ccc} A \otimes \mathbf{Sym}(A) \otimes \mathbf{Sym}(A) & \xrightarrow{\partial_A \otimes 1_{\mathbf{Sym}(A)}} & \mathbf{Sym}(A) \otimes \mathbf{Sym}(A) \\ & \searrow \text{symmetry} & \nearrow 1_{\mathbf{Sym}(A)} \otimes \partial_A \\ & \mathbf{Sym}(A) \otimes A \otimes \mathbf{Sym}(A) & \end{array}$$

References

- [Ehr18] Thomas Ehrhard. An introduction to differential linear logic: proof-nets, models and antiderivatives. *Mathematical Structures in Computer Science*, 28(7):995–1060, 2018.
- [FGHW08] Marcelo Fiore, Nicola Gambino, Martin Hyland, and Glynn Winskel. The cartesian closed bicategory of generalised species of structures. *Journal of the London Mathematical Society*, 77(1):203–220, 2008.